ABSTRACT

I find the QR-Based Smart Attendance System for Office to be a solution that helps companies keep track of employee attendance easily. Old ways such as paper lists, printed sheets and fingerprint machines are slow and hard to use. They can also make mistakes when recording employee attendance. To solve these problems the QR-Based Smart Attendance System for Office uses QR code technology to record attendance. The goal is to make time management simpler and give a way to track employee attendance in office settings.

The system is built with web tools such as HTML, CSS and JavaScript on the side while Python with the Flask Framework works on the back. SQLite Database holds all the data. The front side lets employees and managers use the system without trouble. The back side checks employee details, confirms login, handles QR code scans, and saves attendance records. The database keeps employee details, dates, check-in times and other useful data in an organized way.

In this system every registered employee gets an ID and a QR code. Employees sign in with their username and password. They use a camera to scan their QR code, which marks them as present. When the QR code is scanned, the system automatically writes the attendance data into the SQLite Database. QR code scanning eliminates the need to type and reduces mistakes in employee attendance records. The system also shows attendance reports, employee details and daily summaries on a screen.

Managers can manage employee records, look at attendance logs and study reports. Employees can view their attendance history on their screens. By saving attendance data, the system makes information easier to access and record keeping simpler. It cuts down on paper. The saved data can also help identify attendance patterns, letting managers see if employees are on time and following office rules.

Security and reliability are important parts of the system. User login, employee authentication and QR code checks help determine who can use the attendance tools. QR code scanning alone does not prevent misuse, such as someone pretending to be another employee or sharing a QR code. Combining QR codes with user sessions and server-side checks can make the system safer.

In conclusion, the QR-Based Smart Attendance System for Office provides a practical way to manage employee attendance. By combining QR code technology with web applications and database management, the system reduces manual work, improves accuracy and supports office administration. Future improvements may include mobile app support, instant alerts, cloud integration and advanced reporting tools to meet changing company needs.

Keywords: QR Code Technology, Smart Attendance System, Employee Attendance, Office Management, Web Application, Flask Framework, Python Programming, SQLite Database, Attendance Tracking, Employee Authentication, Digital Record Management, Automated Attendance